

Yichao Separate Training Results: Same-Time B2F and Scalar Future Prediction

OrganoidAgent Yichao analysis

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Abstract

This report covers only the separately trained Yichao models. Stage 1 trained a same-time brightfield-to-fluorescence image model ($B_t \rightarrow F_t$). Stage 2 trained a separate scalar future-expression model from earlier brightfield history and morphology features ($B_{0:k}, X_{0:k} \rightarrow s_{p>k}$). The same-time B2F model contains useful signal, with held-out image MAE 0.1077, expression AUROC 0.728, and fluorescence peak Pearson correlation 0.685. The separate scalar future model does not yet provide useful future prediction, with held-out expression AUROC 0.228 and future peak Pearson correlation 0.134. The practical conclusion from the separate training is that B2F should be kept as a representation-learning task, while scalar future prediction needs stronger temporal targets and a better longitudinal model.

1 Scope

This document intentionally excludes the joint model. It uses only the stage1 and stage2 artifacts:

- Stage 1 B2F model: `\outputroot/stage1_b2f`
- Stage 1 B2F evaluation: `\outputroot/stage1_b2f_evaluation`
- Stage 2 scalar future model: `\outputroot/stage2_future_expression`

2 Dataset

The projected complete/non-edge Yichao dataset was built at:

`\outputroot/future_expression.sqlite`

The B2F stage used 6,691 training crops, 1,601 validation crops, and 1,088 held-out test crops at 256×256 pixels. The scalar future stage used 4,277 training prefix samples, 1,178 validation samples, and 633 held-out test samples. The scalar model used up to five prefix frames and explicit morphology/time features.

3 Separate Model 1: Same-Time B2F

The B2F model learns:

$$\hat{F}_t = G_\theta(B_t), \tag{1}$$

where B_t is a projected brightfield organoid crop and F_t is the same-time projected fluorescence crop.

Table 1: Held-out B2F test metrics from `stage1_b2f_evaluation/test_metrics.json`.

Metric	Value
Test crops	1,088
Expression-labeled crops	14
Image MAE	0.1077
Image MSE	0.0382
Image PSNR	14.18 dB
Expression AUROC	0.728
Average precision	0.1097
Peak Pearson correlation	0.685
Peak MAE	0.746
Best epoch	3
Device	CUDA

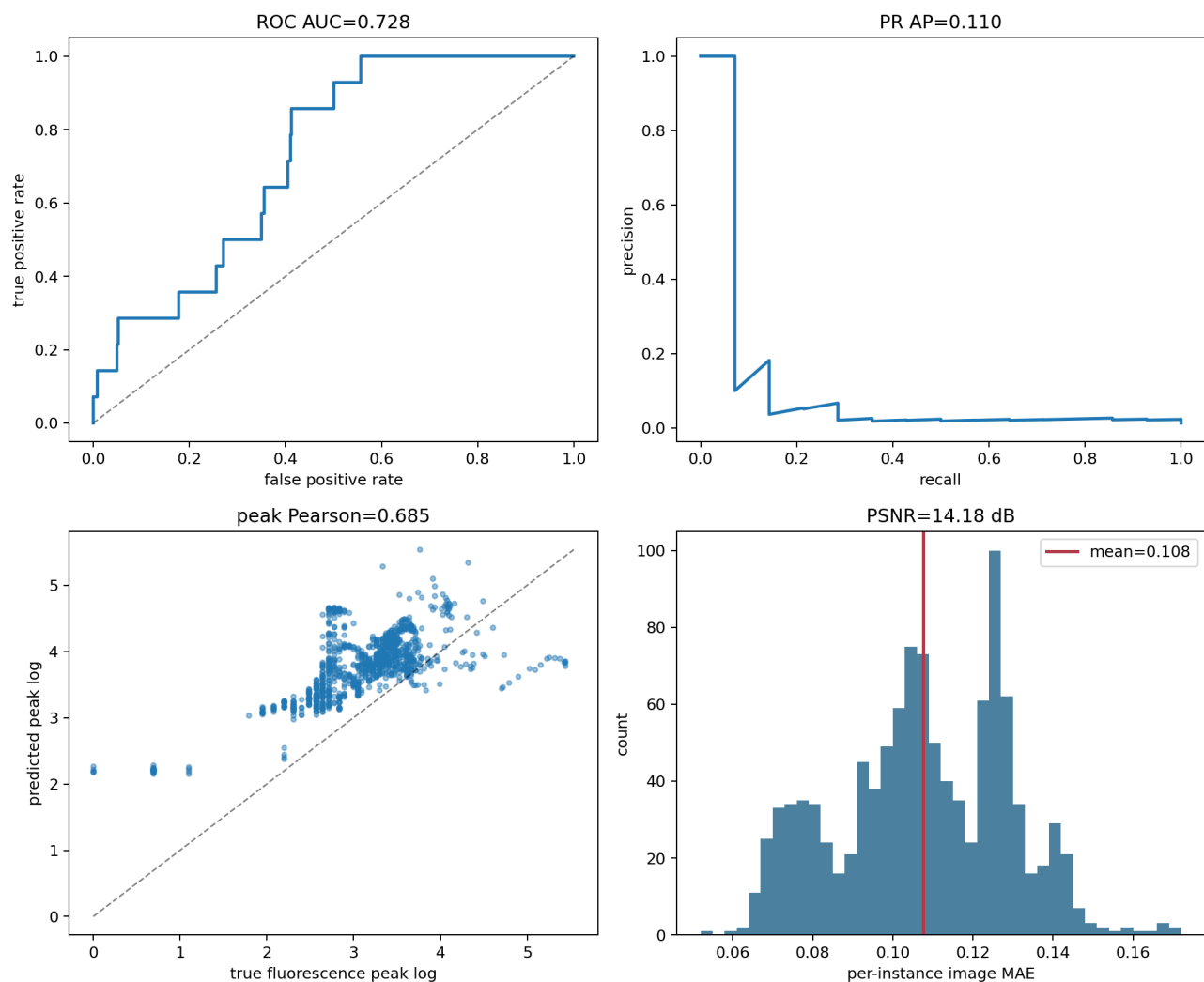


Figure 1: Same-time B2F performance summary from the separate B2F evaluation.

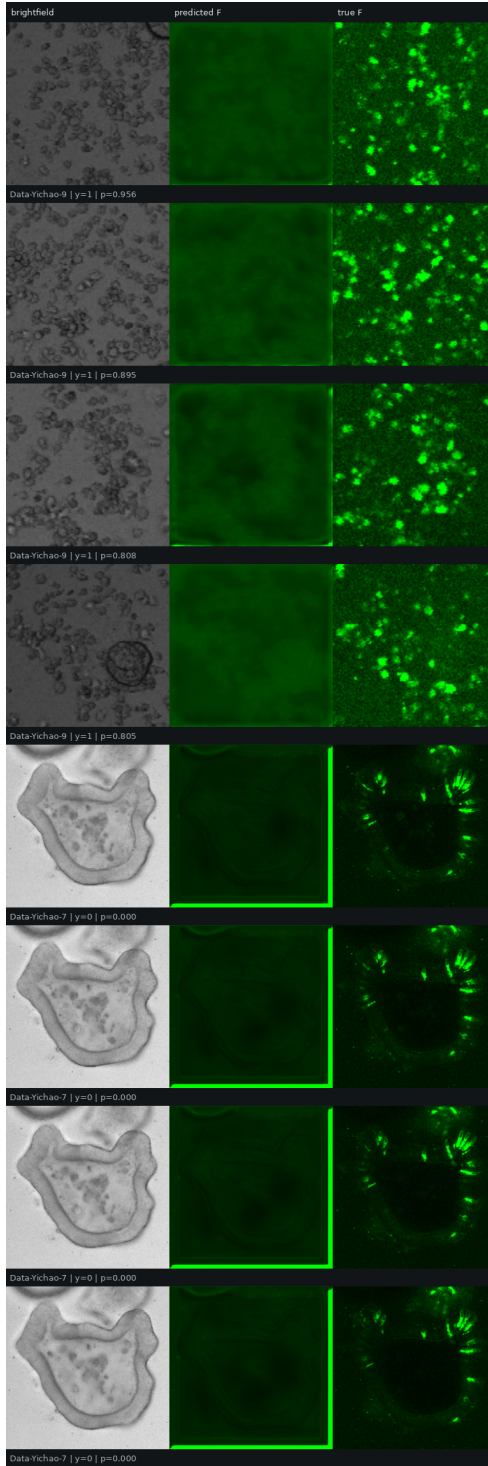


Figure 2: Held-out B2F examples. Columns show brightfield input, predicted fluorescence, and true fluorescence.

4 Separate Feature Explanation

The same-time B2F result suggests that projected brightfield morphology contains some information associated with same-time fluorescence. The explicit feature analysis is useful because it shows which

interpretable morphology variables carry signal.

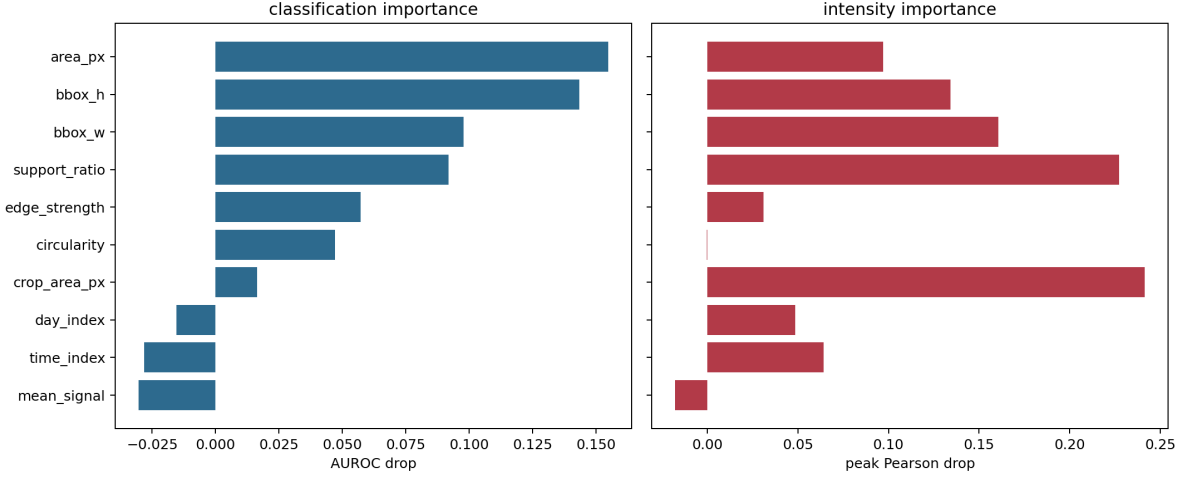


Figure 3: Feature importance summary for same-time fluorescence distinction. Size, bounding-box geometry, support ratio, and edge features are the strongest explanatory variables.

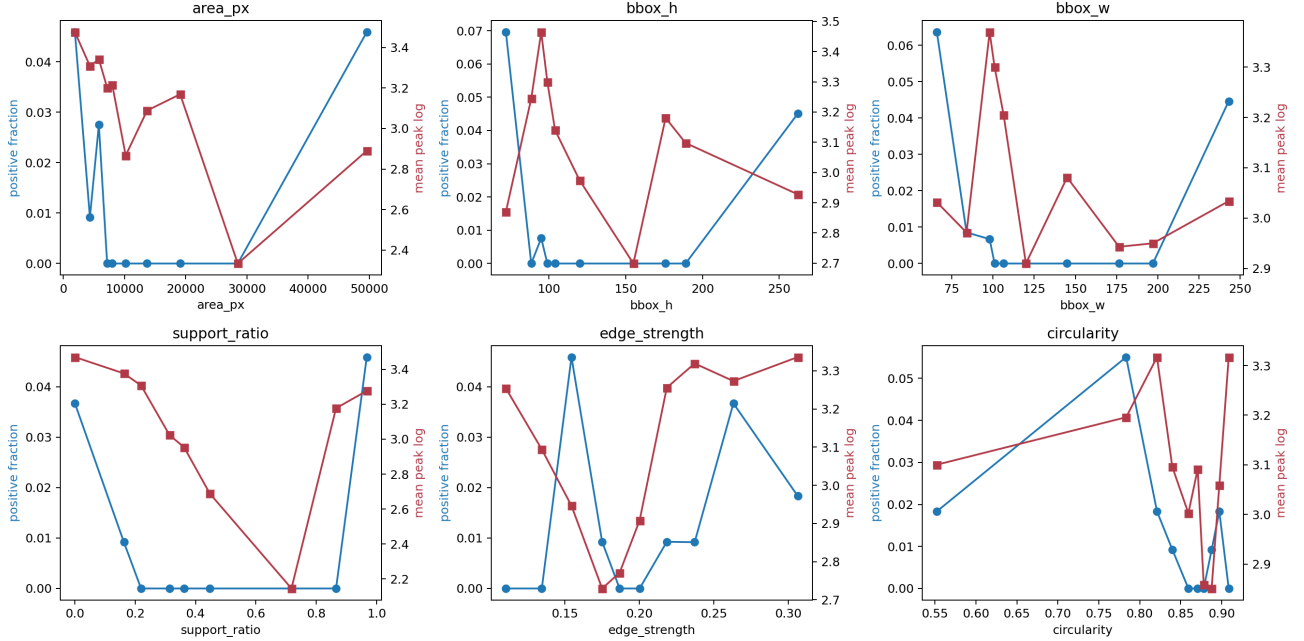


Figure 4: Feature trend plots from the separate B2F evaluation.

5 Separate Model 2: Scalar Future Prediction

The scalar future model takes a prefix of earlier brightfield crops and explicit features:

$$P_k = \{(B_j, X_j)\}_{j=k-L+1}^k, \quad L \leq 5, \quad (2)$$

and predicts future-expression status, future peak fluorescence, and future fluorescence AUC:

$$(\hat{y}_p, \hat{r}_p, \hat{q}_p) = H_\phi(P_k), \quad p > k. \quad (3)$$

Table 2: Held-out scalar future metrics from `stage2_future_expression/test_metrics.json`.

Metric	Value
Test prefix samples	633
Expression AUROC	0.228
Average precision	0.0156
Future peak Pearson correlation	0.134
Future AUC Pearson correlation	0.105
Future peak MAE	0.579
Future AUC MAE	0.929
Image target	Not predicted

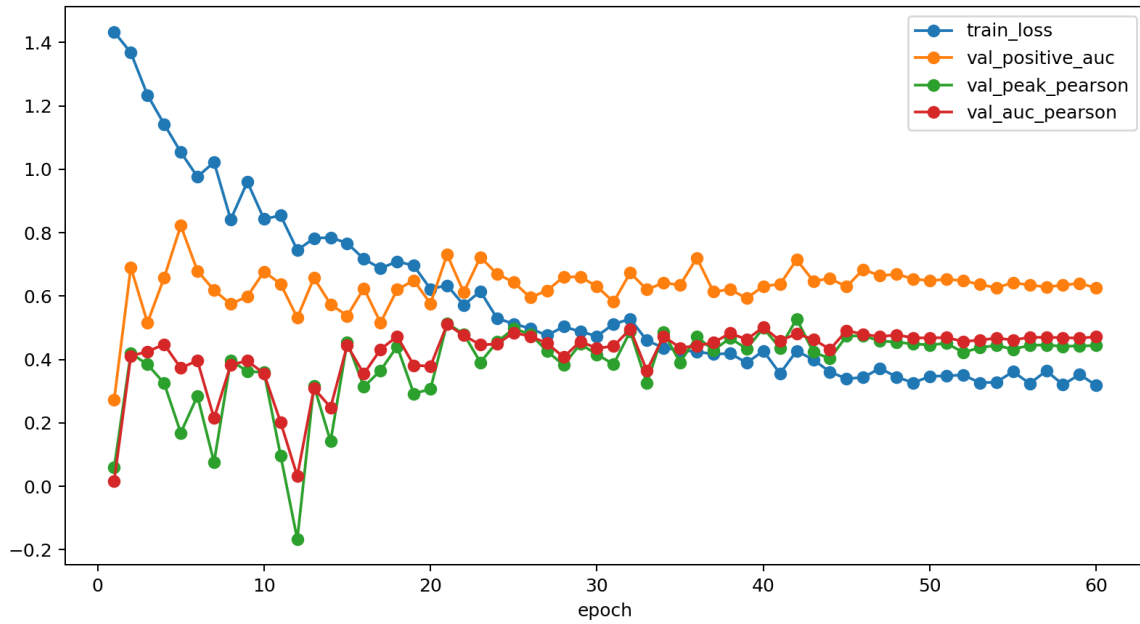
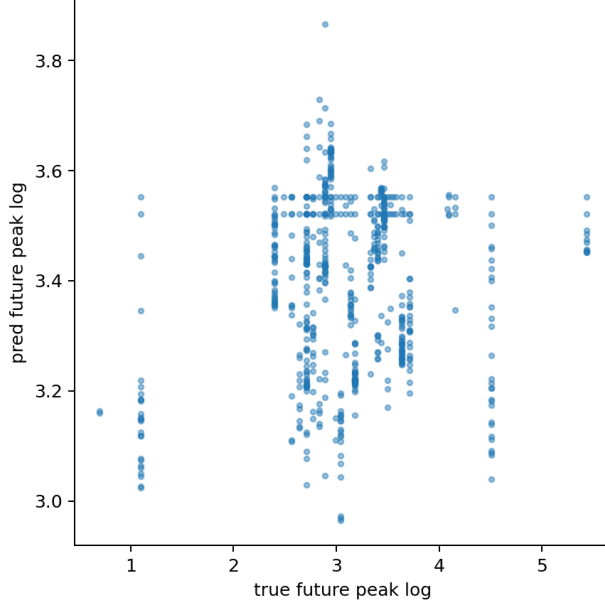
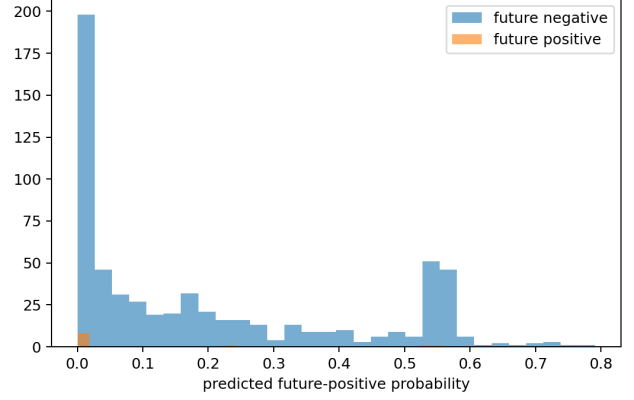


Figure 5: Training curves for the separately trained scalar future model.



(a) Future peak prediction



(b) Future-expression score distribution

Figure 6: Held-out scalar future outputs. The predicted future scalar values are not close enough to the true future values for useful inference.

6 Interpretation

The separate training result is asymmetric. Same-time B2F has measurable signal and should be kept. The separate scalar future model is not sufficient because it discards dense future image structure, relies heavily on approximate track-derived labels, and has too few future-expression examples in the held-out split.

7 Recommended Use

1. Use the B2F model as pretraining for future models rather than treating it as the final biological objective.
2. Do not rely on the separate scalar model alone for future-expression claims.
3. Keep 256×256 as the fast baseline, but test 384×384 or 512×512 because the relevant fluorescent cells can be thin peripheral structures.
4. Improve target construction before scaling model size: verify tracks, use horizon-specific targets, and compare last-future, peak-future, and first-expression-time targets.

8 Conclusion

The separate experiments show that brightfield can predict same-time fluorescence better than chance, but the separately trained scalar future model does not yet solve early future-expression prediction. The strongest next step is to reuse the B2F representation inside a joint temporal model, not to treat the separate scalar future model as sufficient.